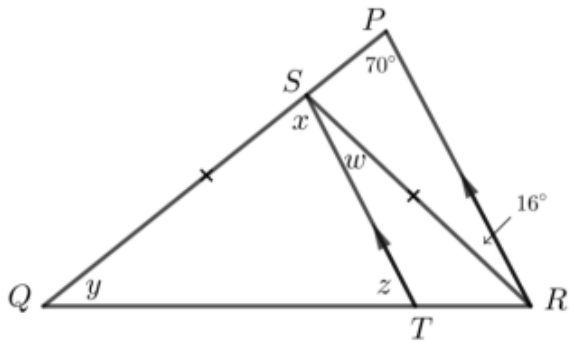
GRADE 10 REVISION: GEOMETRY

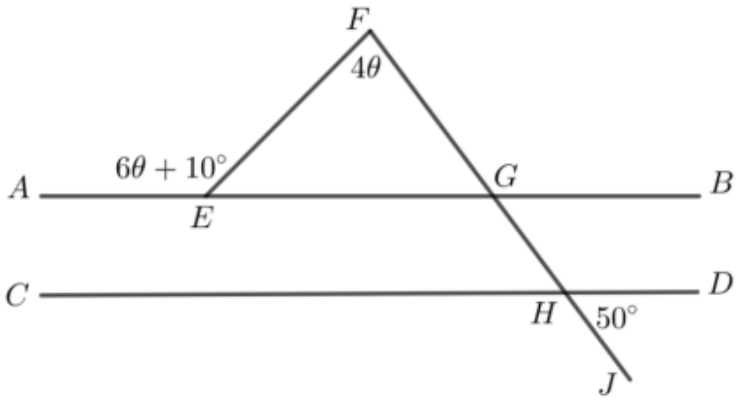
10.2 In the diagram $ST \parallel PR$ and $SR = SQ$. $\hat{P} = 70°$ and $P\hat{R}S = 16°$.



Determine, with reasons, the size of angles w, x, y, and z. (7)

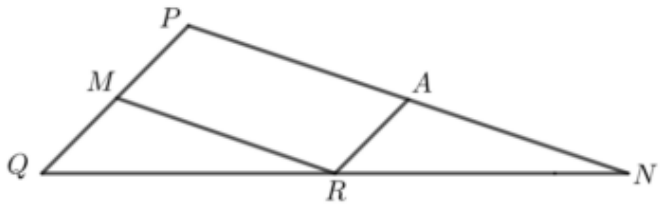
10.3 In the given figure, $AEGB$, $FGHJ$, and CHD are straight lines. $FE = FG$

$A\hat{E}F = 6\theta + 10°$; $\hat{F} = 4\theta$; $J\hat{H}D = 50°$



Prove with full reasoning that $AB \parallel CD$. (7)

10.4 In the figure, $PARM$ is a parallelogram and $QM = PM$.

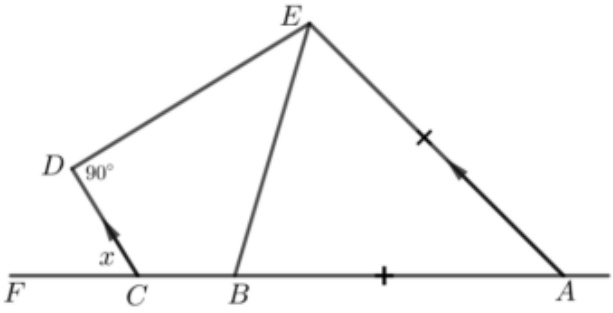


10.4.1 Prove, with reasons, that $\triangle QMR \equiv \triangle QPN$. (4)

10.4.2 Prove, with reasons, that $\triangle NRA \equiv \triangle RQM$. (4)

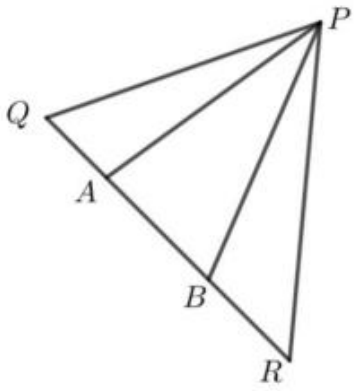
10.4.3 If the perpendicular distance from P to MR is $6cm$ and the area of the parallelogram is $72cm^2$, determine the length of AN. (4)

11.1 In the figure $AE \parallel CD$, $C\hat{D}E = 90°$, $AE = AB$, and $D\hat{C}F = x$



Determine with full reasons, the value of $B\hat{E}D$ in terms of x. (5)

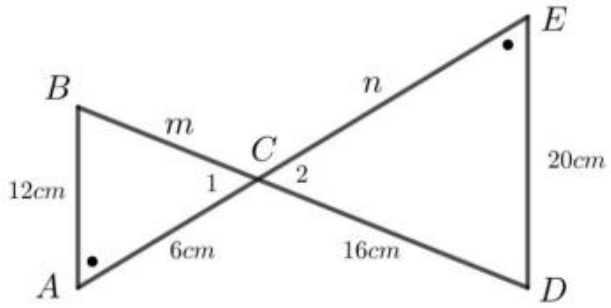
11.4 In the figure, A and B lie on side QR of ΔPQR such that $QA = RB$. $PQ = PR$.



11.4.1 Prove, with reasons, that $\Delta APQ \equiv \Delta BPR$. (4)

11.4.2 Prove, with reasons, that $P\hat{A}B = P\hat{B}A$. (3)

11.5 In the figure, $\hat{A} = \hat{E}$.



11.5.1 Prove that $\Delta ABC \;|||\; \Delta EDC$. (4)

11.5.2 Hence determine the values of m and n. (4)